National College of Ireland

Project Submission Sheet

Student Name: ……Muddassir Ahmed ……

Student ID: …… x23138688 ……

Programme: ……MSC Data Analytics…… **Year:** ……2024……

Module: ……Data Mining & Machine Learning 2 ……

Lecturer: …… ……

Submission Due Date: ……19/05/2024 ……

Project Title: ……Case Study & Paper Review ……

Word Count: ……4181 ……

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.

ALL internet material must be referenced in the references section. Students are encouraged to use the Harvard Referencing Standard supplied by the Library. To use other author's written or electronic work is illegal (plagiarism) and may result in disciplinary action. Students may be required to undergo a viva (oral examination) if there is suspicion about the validity of their submitted work.

Signature: ……Muddassir Ahmed……

Date: ……19/05/2024……

PLEASE READ THE FOLLOWING INSTRUCTIONS:

1. Please attach a completed copy of this sheet to each project (including multiple copies).
2. Projects should be submitted to your Programme Coordinator.
3. **You must ensure that you retain a HARD COPY of ALL projects**, both for your own reference and in case a project is lost or mislaid. It is not sufficient to keep a copy on computer. Please do not bind projects or place in covers unless specifically requested.
4. You must ensure that all projects are submitted to your Programme Coordinator on or before the required submission date. **Late submissions will incur penalties.**
5. All projects must be submitted and passed in order to successfully complete the year. **Any project/assignment not submitted will be marked as a fail.**

Office Use Only	
Signature:	
Date:	
Penalty Applied (if applicable):	

AI Acknowledgement Supplement

[DOMAIN APPLICATION PREDICTIVE ANALYTICS]

[A PREDICTIVE ANALYTICS APPROACH FOR FINANCIAL INSTITUTIONS OF USA BY USING INCOME DATASET]

Your Name/Student Number	Course	Date
Muddassir Ahmed	MSC Data Analytics	18/05/2024

AI ACKNOWLEDGMENT

Tool Name	Brief Description	Link to tool
N/A	N/A	

Question 1 - Case Study

Prediction of volcanic eruption

Introduction

The field of geoscience presents a pertinent challenge: the anticipation of volcanic eruptions carries significant importance for the safety of human beings and the conservation of the environment. This report uses data analytics and machine learning to develop a methodology for the anticipation of volcanic deterioration. Through exploratory data analysis and feature engineering and model selection, we create a method to identify relevant information from datasets that include numerous variables from the seismic and geological-field as well as the atmosphere. In addition to the concern of scalability in predicting future dangers, this proposition also provides for an ethical analysis that by portraying the impacts and importance of responsible and transparent decisions.

1. Exploratory Data Analysis / Data Cleaning

For the domain of volcanic eruption prediction, exploratory data analysis is the first step in the methodology we propose, as well as the extensive data cleaning. The underlying assumption is that the dataset features various information inputs, such as seismic data, gas emissions, and geological parameters, providing a summary of volcanic behaviour. (Carranza, E.J.M., 2023) EDA encompasses descriptive statistical techniques and various visualizations to determine the nature of data distributions, identify outliers, and evaluate missing values. In recent literature, suggested techniques for missing data are mean imputation or interpolation, while for outliers robust statistical techniques or domain-knowledge based methods are used. Additionally, data cleaning refers to inconsistencies or errors in data entry, which ensures data integrity with subsequent analysis. Difficulties may occur when integrating and handling different data sources or dealing with data quality issues, thus requiring thoughtful consideration and perhaps iterative refinement of cleaning methodology. Nonetheless, these challenges are crucially important to address as they ensure data is of high quality to proceed with subsequent stages of the methodology leading to meaningful results and reliable model building for volcanic eruption prediction.

2. Dimensionality Reduction / Feature Selection

Further concerning the methodology of volcanic eruption prediction, the subsequent step essential to take is dimensionality reduction and feature selection. The assumptions are that data is high-dimensional, that is, it contains a large number of features, which may entail the computational inefficiency and the curse of dimensionality. The existing literature like (Cárdenas-Peña, D.,G., 2013.) calls for the implementation of the dimensionality reduction techniques to address these issues and ensure the better interpretability and generalization of the model presented. The primary method we choose for the dimensionality reduction is the Principal Component Analysis. It is preferred due to the fact that it defines the linear transformation of data in a way to obtain the maximum variance while losing the minimum amount of information. Specifically, PCA transforms the feature space of the original measurements into a lower-dimensional subspace of uncorrelated principal components, which

is more computationally efficient due to the lack of correlations between dimensions and easier for interpretation.

Additionally, feature selection is essential for identifying the best-informative features for volcanic eruption prediction and discarding redundancy and noise. We use the method of Recursive Feature Elimination to find only the most discriminative features and also utilize a feature importance score using multiple models to rank the importance of features. RFE discards the less important one based on their performance and retains the important ones that contain the required info for prediction tasks. Ensemble-based method feature importance ranking employs multiple intelligence to rank the importance of each feature sync efficiently.

Therefore, in our work, we do not consider simple approaches, such as manual selection of features or methods based on thresholds, as they entail numerous cases of missing subtle patterns that have great value in the data. In addition, we exclude methods for dimensionality reduction, such as Linear Discriminant Analysis, as they are more suitable for classification problems and may not properly describe the data's intrinsic structure. Also, it is definitely tough to establish the most appropriate number of principal components or to maintain a balance between the importance of the feature and the computational complexity. It is of high importance to conduct numerous experiments and validations so that the selected features play a meaningful role in the model's predictability. We strive to simplify the model-building process and improve the predictive power of predicting volcanic eruptions through a thorough grasp of dimensionality reduction and feature selection.

3. Feature Engineering / Feature Extraction

Feature engineering and extraction are essential to empower the models to predict, enhancing the predictive capacity of the models by discovering new insights and collecting relevant information from the raw data source. The key assumptions are data availability in a wide variety of types, such as seismic signals, gas emissions, and geological features. Each offers a distinctive view of the data and influences the volcano's behaviour. Recent works like (Asker, L. and Maclin, R., 1997) in literature use feature engineering specific to the domain's characteristics. This gives the opportunity to create features which are informative and that capture the spatial-temporal relationships between volcanic phenomena. We propose the use of volcanology and geophysics knowledge-driven feature engineering methods to create features that encode encounters between instances and patterns.

In order to capture the temporal dynamics or frequency characteristics of the aforementioned seismic signals, other time-series data, feature extraction techniques such as wavelet transforms or Fourier analysis would be appropriate. The methodology would allow extracting the spectral energy distribution or signal entropy, which while both are discriminative features representing the state of volcanic activity in many cases, to our belief. We are also against simple feature extractions with rough estimations based on raw sensor data or elementary statistical measures, which can omit patterns in the data. We are also not in Favor of other analytically powerful feature extractions that can be both computationally expensive and partially non-interpretable.

A major challenge is choosing the appropriate combination of feature engineering methods for effectively capturing the complex nature of volcanic systems without problems of data overfitting and sparsity. Furthermore, as large volumes of seismic data are processed, it is important to optimize the scalability and computational efficiency of the feature extraction

process. To overcome these challenges, feature engineering methods must be chosen carefully and thoroughly validated to determine their impact on the overall model performance and forecasting accuracy in predicting eruptions. By employing feature engineering and extraction carefully, our methodology seeks to make the most of the latent knowledge present in heterogeneous datasets to produce more dependable and trustworthy predictive models for volcanic hazard evaluation.

4. Choice of modelling techniques

The selection of modelling techniques is essential to accurate and reliable deployment of volcanic eruption prediction. It is important to recognize that during this step of the methodology, it is assumed that necessary preprocessing has already taken place and that the data includes the selected predictors. Recent advances in the scientific literature paper by (Kavzoglu, T. and Teke, A., 2022) show that machine-learning models are highly effective at learning the complex relationships contained within volcanic data for precise event prediction. We propose an approach for the utilization of ensemble learning methods. Specifically, we support the employment of Gradient Boosting Machines. GBMs are uniquely well-suited to handle the multiple types of heterogeneity and complex nonlinearity found in this type of data, as well as the frequent issue of imbalanced data often found in volcanic system monitoring.

GBM, such as XGBoost or LightGBM, have the following properties that make them a good fit for volcanic eruption prediction tasks: robustness, scalability, and interpretability. These models are built and trained in an iterative fashion, wherein the outputs of several weak learners are combined to improve the prediction. This structure guards the model against overfitting and enhances the generalization abilities. Moreover, GBM models do not tend to require imputation or removal of missing data and outliers, and they often do not necessitate more data cleaning and preprocessing. Multidimensional data, with numerical, categorical, and spatiotemporal features, are preserved in their original forms. Conversely, linear regression or decision trees cannot model the nonlinearity and temporal dependence structure in volcanic data and are hence ruled out. Deep learning, like LSTM networks, can be used but often overfits due to the number of data points provided and will, therefore, require an excessive number of computations and is not suggested.

The primary challenges to address include the optimization of hyperparameters and class imbalance. Eruption prediction datasets are typically heavily imbalanced, with eruption events being extremely rare in comparison to non-eruption periods. Model interpretability is also an ongoing issue, especially when it comes to deploying models to assist in decision-making for volcanic hazard reduction. Overall, addressing these challenges requires a lot of experimentation, validation, and model tuning to achieve the desired performance and robustness. Utilizing ensemble learning approaches such as GBM, our method can be used to develop accurate and robust models for eruption forecasting. Combining the latest developments in machine learning algorithms with the traditional feature engineering based on domain and expert knowledge help to improve the forecasting accuracy which is beneficial in contributing to the better assessment and mitigation of volcanic hazards.

5. Hyperparameter Optimisation

When it comes to predicting volcanic eruptions, hyperparameter optimization is essential for optimizing machine learning models and achieving the optimal predictiveness. The general

assumptions are that an actual evaluation metric, such as accuracy or F1-score, is established to measure the model performance of the mint, and so this is available throughout the optimization process. Current research suggests that this broad category of optimization model, especially in high-dimensional spaces or optimization problems with non-smooth, noisy, or non-convex objective functions, represented by Bayesian Optimization for Hyperparameter Tuning, or BOHT, directs the literature.

BOHT utilizes probabilistic models to approximate the objective and. We reiterate the procedure of selecting hyperparameters based on their possible evaluations to maximize the expected improvement in the model's performance. Drawing an analogy to kernel density estimation, BOHT redefines the problem we would like to solve within the parameter space, shortening it to the query which has not been previously evaluated. The application of sampling, combined with the previously explored evaluations, allows the model to converge to the optimal hyperparameter configuration without making an exhaustive search through the grid or randomly. This way, it proscribes any computational burden connected to the grid or random search. Hence, we denounce trivial optimization techniques such as grid describe by (Oláh, L. and Tanaka, H.K., 2022) or random searches due to their complexity in exploration of the parameter space and obtain maximum values. The high-dimensional nature of the problems is dependent on the limited automated methodology because they work nominal dimensions. Fallowing the precedence of the evolutionary methodology, such as genetic algorithms, may have the poor outcome while converging slowly with a noisy target.

As optimizing computational hyperparameters can involve difficulties in specifying optimal hyperparameter search spaces and the appropriate exploration-exploitation balancing, this is of increasing importance. There is also the challenge of obtaining optimization results that can be repeated precisely across various additional analyses of a model so that prior trust in model performance can be sustained. To that end, our proposed methodology employing BOHT and other Bayesian optimization processes will attempt to guide the hunt for optimal hyperparameters to allow much better machine-learning models for volcanological estimating of volcanic eruptions. Thus, we will be able to increase the robustness of our predictive models and improve their reliability by not scrambling their hyperparameter values and not use infeasible configurations. Subsequently, our models will allow the timely assessment of volcanic hazards.

6. Model Evaluation

Finally, it is essential to consider model evaluation concerning volcanic eruption prediction. Prior models treated assumptions that. First, all models should have sufficient labelled data to train and evaluate such models, as shown in drawings representing eruption and non-eruption periods. Second, comprehensive evaluation metrics such as accuracy, precision, recall, and F1-score have been used to evaluate models' performance regarding eruption aspect. Lastly, cross-validation should be considered to avoid overfitting during model assessment. Cross-validation splits the dataset into various folds during training and testing. Hence, it generates reliable estimates regarding the developed models on unseen new data. We avoid simple evaluation strategies, such as single train-test splits, which introduce bias into the performance metrics and may easily exaggerate how good the model is at predicting eruptions. We also face issues with imbalanced classes and selecting appropriate metrics that adequately reflect the model's performance regarding the decisions to be made based on eruption predictions, which requires

special attention and extensive validation efforts. In doing so, we believe our approach to model evaluation supports the accurate and valid evaluation of predictions that guide the development of strategies to mitigate volcanic threats.

7. Scalability Issues

One of the major assumptions regarding the challenge of scalability is that the solutions should include assumptions on the efficient processing and analysis of streaming seismic sensor data and satellite imagery which is available from Realtime observational systems. Studied literature further revealed that the potential implementation of scalable computing frameworks, such as Apache Spark, and distributed machine learning models may respond to the challenge of big data in volcano monitoring. The proponents argue that distributed computing solutions like Apache Spark are recommended for the analysis of large-scale volcano datasets as they operate on parallel processing and are "fault-tolerant for analyzing large-volume data". This means that the processes of giant-scale dataset computations can be distributed as they can run on more than one node. Traditional single-node computing approaches are insufficient due to limited scalability. Volcanic monitoring systems constantly produce a high volume and velocity of data, which cannot be effectively processed in real-time using traditional computing methods. Additionally, services such as Amazon Web Services or Google Cloud Platform provide scalability and elasticity, allowing for the dynamic allocation of computing resources based on demand. Although scalability and elasticity-related configurations are beyond the scope of this work, challenges may also arise in ensuring the performance optimization of distributed computing frameworks, as well as the consistency and integrity of the data in distributed environments. Furthermore, the deployment and maintenance of distributed systems may be more complex and require additional expertise, which increases the operational overhead.

8. Ethical implications

The ethics of prediction surrounding volcanic eruption primarily concern false alarms or failed predictions, which can cause mass hysteria or neglect among communities. The following assumptions will be made: the decision-making processes must be transparent and held accountable, stakeholders and domain experts must be engaged in model development and validation. Given the most recent papers, it is evident that careful prejudice elimination and fairness when it comes to data collection and model predictions along with vulnerable populations in volcanic hazard area should also be taken into account. It will be critical to recognize that these ethical considerations must be met in a way that acknowledges the dichotomy between the urgency for public safety and the responsible and ethical use of predictive modelling in volcanic hazard mitigation efforts.

Conclusion

The proposed methodology for volcanic eruption prediction involves several stages, all of which draw on recent literature and are uniquely designed to mitigate each challenge in the domain. This exploration of the data is followed by the stage of data cleaning, which ensures the integrity of the data and discovers patterns originating from several sources. The dimensionality and feature reduction are designed to compact the data; nonetheless, the retained information is still essential. Feature selection identifies the distinctive patterns originating from the data, ensuring high-quality predictions. The ensemble models, specifically

the Gradient Boosting Machines, are chosen for the strong performance and interpretable nature of the model. Hyperparameter optimization chooses optimal values for the spatial models produce optimal results while saving the computational power. The final evaluation of model accuracy with the help of the cross-validation methods tackles the overfitting and bias. The issue of scalability is resolved through the utilization of the framework for distributed computing, in this case, Apache Spark.

References

Voight, B., 1988. A method for prediction of volcanic eruptions. Nature, 332(6160), pp.125-130.

Carranza, E.J.M., 2023. Exploratory data analysis. In Encyclopedia of Mathematical Geosciences (pp. 364-368). Cham: Springer International Publishing.

Cárdenas-Peña, D., Orozco-Alzate, M. and Castellanos-Dominguez, G., 2013. Selection of time-variant features for earthquake classification at the Nevado-del-Ruiz volcano. Computers & geosciences, 51, pp.293-304.

Asker, L. and Maclin, R., 1997, July. Feature engineering and classi er selection: a case study in Venusian volcano detection. In Proceedings of the Fourteenth International Conference on Machine Learning (pp. 3-11).

Oláh, L. and Tanaka, H.K., 2022. Machine learning with muographic images as input: An application to volcano eruption forecasting. Muography: Exploring Earth's Subsurface with Elementary Particles, pp.43-54.

Question 2 - Paper Review

Title: A machine learning approach for hierarchical classification of software requirements

Structure & Title: It gives better understanding of the paper, all words used in the title gives overall meaning of the paper. Any person can understand that in this paper machine learning approaches are used for classification problem of software engineering.

Abstract: This paper address machine learning approach for software requirement classification. It present HC4RC approach which is addressing challenges such as high dimensionality with small data size that is appearing in software requirement classification. In this paper semantic is use for feature selection and it is identifying functional and non-functional requirements. Classification framework is use to enhance performance after collecting the different requirements. After performing experiments, it is showing HC4RC approach is well for classifying software requirement, which is showcasing its potential for real world applications in software engineering.

Introduction Review: This part explains about Machine learning (ML) approaches that is use to increase results in different real world applications and it is showing potential in different tasks. These task can be fake news detection, sentiment analysis and medical diagnosis. In this requirement engineering research, Machine learning techniques are used for requirement classification, prioritization and traceability. Requirement classification play very import role in this research. It is also use here in the organization and understanding of diverse requirement.

It highlights some hurdles for applying machine learning algorithms due to class imbalance and the high dimensionality, with the low sample size data. To resolve this main issue, HC4RC approach is introduced in this paper for better solution. A machine learning framework was tailored for classification of software requirements. Semantic is used for dataset decomposition, feature selection and hierarchical classification. HC4RC approach is use to enhance the performance of the classification in the availability of imbalance class and limited dataset.

This introduction part is highlight the significance of requirement classification. HC4RC approach use to overcome the software engineering challenges. Some complexities and limitations are highlighted in HC4Rc approach and its gives solutions for improving classification accuracy and efficiency in software requirements.

Highlights:

1. Addressing Critical Challenges: HC4RC is well tackling the issues of class imbalance and HDLSS that is ultimately have impact on machine learning performance methods in software requirement classification.
2. Innovative Approach: This paper introduce technique including semantic role based for feature selection and dataset decomposition and hierarchical classification to overcome software requirement challenges.
3. Comparative Analysis: Results showing HC4RC gives best results as compare to other existing approaches and it is showcasing its effectiveness for handling imbalance different classes and HDLSS dataset.

4. Practical Contribution: HC4RC gives best practical solution for real world challenges in software requirement classification and it is enhancing the reliability and efficiency of machine learning.

Methodology Review: HC4RC approach is used to tackle software requirement and evaluation problems and hierarchical classification of software requirements. Proposed approach address two critical challenges in requirement classification. One of them is class imbalance and other one is high dimensionality, with the low sample size data. To resolve these two challenges, HC4RC perform semantic roles based feature selection, dataset decomposition and the hierarchical classification.

Semantic Role-based Feature Selection: This is used for small numbers of semantic roles, just to identify the most relevant features. They get features from the requirements to address the dimensionality, and low sample size.

Dataset Decomposition: This type of technique is used to rebalance dataset, which is divided into two datasets, one is consist of majority classes and other is with minority classes.

Hierarchical Classification: Researcher used for decomposition of the dataset, and it is performing hierarchical classification on decomposed dataset.

- Text pre-processing: It is used to clean and standardize text, because text consist of different unwanted words. It is also highlight natural NLP are available for the text pre-processing. In HC4RC, NPL techniques were implemented for text, tokenization and lowercasing and short words removal.
- Semantic role-based feature selection: In this step SR4Fs feature selection technique was implemented. It is also explaining role of semantic. Semantic role in words plays a certain role (Gildea & Jurafsky, 2002). In this research only six points was used because these are similar to this paper requirement (Rolland & Proix, 1992).

Six semantic roles are mentioned below:

1. Agent – It is explained as volitional causer of an event or action (Jurafsky & Martin, 2020). It paly very important role here.

2. Action – This is a state and this role explain verb of sentence. For example, when any one login to any new computer, system will generate email to user for verification.

3. Theme – Any person effected by any action. (Jurafsky & Martin, 2020). For example, system will send message to user for verification of unfamiliar computer. Message will be act as a theme. (Sutcliffe & Maiden, 1998).

4. Goal – destination of the action is called goal. (Jurafsky & Martin, 2020). For example, system will send message to user for verification of unfamiliar computer. Word user is consider as a goal here.

5. Manner – action take place in which environment is called manner. (Xue, 2008). For example, system should be very easy to user.

6. Measure – action is control by which degree and event is explained in this section. (Jurafsky & Martin, 2020). For example, system remain available for user 98% in working days.

Results: In this paper result is explained in four steps. Comparison, analysis and interpretation is discussed in this section. It is also explaining why their approach is better than other approaches.

1. Comparing of HC4RC with K&M

It is explained that HC4RC and K&M are SVM model based and they have achieved better performance by including the semantic role based feature selection. They have decomposed dataset and handled imbalanced classes are better than K&M approach. They show HC4Rc has high macro and micro average as compare to K&M. Results also shows HC4RC gives better generalizability on all unseen software requirements.

2. Comparing HC4RC with Yin

It highlights both HC4Rc and Yin useful decomposition to address class imbalance issue but HC4Rc gives better performance. In multi classification HC4RC gives better results as compare to Yin approach. Its shows HC4RC gives high macro and micro average than Yin. Results also shows HC4RC giving better generalizability on all unseen software requirements.

3. Comparing HC4RC with NoRBERT

They have noticed NoRBERT performance varies in small classes but NoRBERT gives bad performance on Maintainability class as compare to HC4RC. NoRBERT provide poor performanc. Results also shows NoRBERT giving the better generalizability on all unseen software requirements.

4. Further analysis and discussion

It highlights NoRBERT give best for requirement classification. HC4RC overall gives overall best results for requirement classification. HC4RC has reached the best performance on all individual classes.

Conclusion: Researcher claim that HC4RC is best machine learning approach for multiclass classification of software requirements. They highlight two main points of this problem. These are common problem in requirement classification. First problem is to resolved through dataset breakdown and the hierarchical classification. Second problem deal with semantic role based feature selection. Researcher highlight HC4RC perform better as compare to SVM and K&M and Yin approach. Furthermore, researcher claim their semantic role base feature selection method provide more effective solution to class imbalance. Their findings highlights HC4RC perform better than all other approaches and gives better generalizability. It highlights multiclass classification in class imbalance and HDLSS is the challenge for machine learning approach. For future work Suggestion, it highlights more work is required to develop the better feature selection, for example semantic and different roles of the software requirements. Researcher use only one dataset and it suggest machine learning is based on the outdated classification models and more work is required to improved feature selection for semantic roles. They believe approaches they have applied gives better results and performance as compare to others.

Language: Structure of the paper was well and it explained all points very briefly with the use of proper diagrams. Diagrams are clearly defined in each section and proper labeling is perform. While reading the abstract part we can know all paper methodology and its outcomes. There was no spelling and grammar mistakes found in overall paper. Each step starts with proper heading and numbering.

Previous Research: Previous machine learning models and their comparing is well elaborated in this paper. In literature reviews part reference are available with explanation. Researcher highlights problems which are exist in previous machine learning models with reference.